

AEP-X Ultra — Agentic Development Life Cycle (ADLC) Plan

Autonomous Enterprise Protocol eXtended — Universal Operating System for Agentic AI

Source AEP-X Ultra Project Manual — Consolidated Edition (3,188 pages) Prepared 7 July 2026 Status Draft v1.0 for review

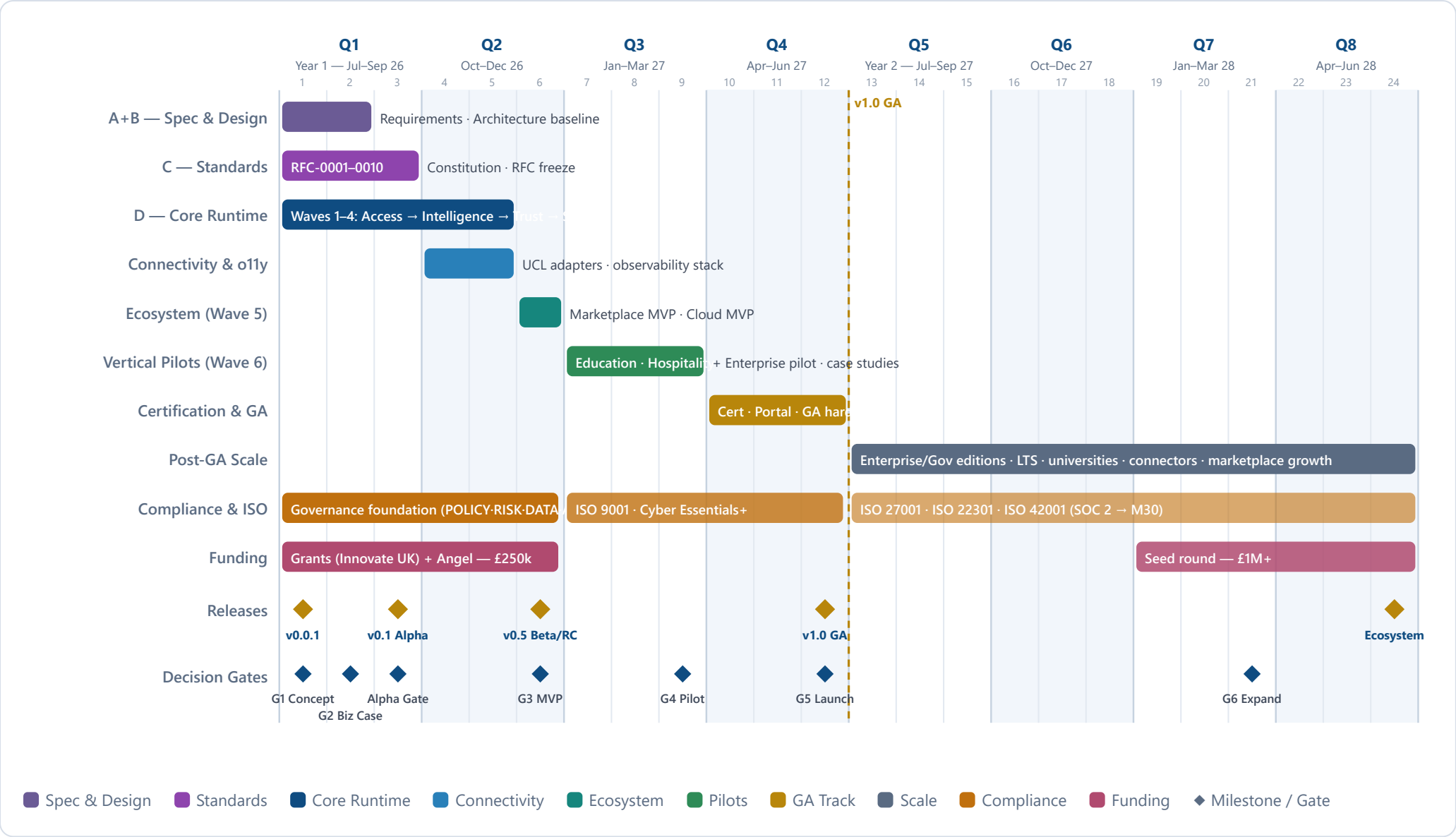
Owner Equality Software Ltd / AEP-X Programme

1. Purpose & Scope

This ADLC Plan translates the AEP-X Ultra Project Manual into a single, executable development life cycle. The manual contains many overlapping roadmaps (Stages 1–20, Phases 1–20, G1–G10, 12 workstreams, 100-day plans, 24-month plans, 5-year and 10-year programmes). This plan **reconciles them into one lifecycle** with clear phases, gates, deliverables, and metrics, following the manual's own directive: "*stop creating stages, start shipping*" and the 180-day Execution Lock (no new architecture, no new framework layers, no new conceptual stages).

What is being built: AEP-X Ultra — the Autonomous Enterprise Protocol eXtended, a universal interoperability layer and runtime for agentic AI: cache-first cost optimisation, multi-layer memory, anti-hallucination/evidence engine, trust & governance, universal connectivity (MCP, REST, gRPC, Kafka, etc.), marketplace, and certification ecosystem.

Roadmap at a Glance (24 Months)



Waves 1–4 overlap by design: identity/trust are stubbed in Month 1 and hardened in Month 5 (§6.1). The compliance track continues to Month 30 (SOC 2) beyond this chart. Detailed schedule in §6.2; gate definitions in §7 and §11.

2. ADLC Methodology

The lifecycle is **SPARC-X native** (the manual's own methodology), wrapped around continuous agentic loops:

ADLC Phase	SPARC-X	Core Question
A. Specification	S — Specification	What must the system do, for whom, under which constraints?
B. Design	P + A — Pseudocode, Architecture	How do agents reason, remember, decide, recover?
C. Standards & Governance	(Constitution/RFC track)	What rules bind every component before code exists?
D. Implementation	(Build waves)	Build the 13 core services + SDKs, spec-driven
E. Validation	R — Refinement	Does it meet the gates: performance, safety, cost, compliance?
F. Release	C — Completion	Alpha → Beta → RC → GA with exit criteria
G. Operations	X — Continuous Optimisation	Run, observe, respond, comply
H. Learning & Evolution	X — Continuous Optimisation	Feedback → Learning → Model/RFC updates (the agentic loop)

Two invariant loops run through every phase:

- **Anti-hallucination loop** (RFC-0007): Question → Memory → Knowledge → Evidence → Validation → Response, with confidence bands (Verified 95–100 / High 80–94 / Medium 60–79 / Unverified 0–59) and mandatory human oversight for S3/S4 risk classes.

- **Cost-elimination loop** (RFC-0008): Cache → Memory → Knowledge → Workflow → Tool → Prediction → Small → Medium → Large Model → Human. The LLM is the last resort. (Invariants C1–C4: never invoke an LLM if cache/memory/knowledge has the answer; never use a premium model when a cheaper one meets confidence thresholds.)

And ten constitutional laws govern all development (AEP-X Zero), most notably: *Identity Before Interaction, Trust Before Execution, Evidence Before Assertion, Knowledge Before Generation, Reuse Before Computation, Human Authority in critical domains* (finance, legal, healthcare, child-related decisions).

3. Phase A — Specification (Weeks 1–4)

Objective: Freeze what is being built and why, before any architecture work.

Deliverable	Content	Source
Vision & Strategy Pack	Executive vision, PESTEL analysis, SWOT, mission KPIs	Manual pp. 4–6
Business Requirements (BRD)	Stakeholder analysis, market gap, competitive positioning (vs MCP, LangGraph, CrewAI, AutoGen)	PRD-001
Functional Requirements	FR-001...FR-013 — one per core service (Registry, Discovery, Gateway, Memory, Cache, Knowledge, Identity, Trust, Policy, Audit, Workflow, Safety, Verification)	PRD-001
Non-Functional Requirements	Performance, availability, scalability (10M agents design ceiling), security, cost targets (\$9)	Phase 3 spec
Compliance Requirements	UK GDPR, GDPR, EU AI Act, ISO 27001, ISO 42001, SOC 2, NIST AI RMF, DPA 2018	SEC-001, AIA-001
User Stories & Acceptance Criteria	100–500 stories, product backlog	Phase 2 spec

Gate A (Concept Approval): Strategic alignment confirmed; requirements baselined; risk register opened.

4. Phase B — Design (Weeks 3–8, overlaps A)

Objective: Produce the architecture baseline that all code is generated from.

Deliverables:

1. **Domain-Driven Design** — 8 core domains, bounded contexts.

2. **Protocol Design** — message envelope, 8 message types, capability/agent/cost/safety contracts.
3. **Layered Reference Architecture** — the 20-layer model (Infrastructure → Connectivity → ... → Constitution) mapped to microservices.
4. **Data Architecture** — canonical data model; PostgreSQL (system of record), pgvector (embeddings), Neo4j (knowledge graph), Redis (cache), Kafka (events), MinIO (objects).
5. **Memory Design** — M0 Prompt → M1 Session → M2 Episodic → M3 Semantic → M4 Procedural → M5 Organisational → M6 Federated, each with TTL and governance metadata.
6. **Cache Design** — L0 Prompt (5 min) → L1 Session (1 h) → L2 Agent (24 h) → L3 Workflow (7 d) → L4 Organisation (30 d) → L5 Federation (policy), target >90% hit rate.
7. **Reasoning/Decision/Risk/Recovery flows** (SPARC-P pseudocode) for every agent and the workflow engine.
8. **Security Architecture** — Zero Trust; OIDC/OAuth2/JWT/mTLS; RBAC + ABAC + PBAC; TLS 1.3 in transit, AES-256 at rest; Vault for secrets; threat model covering prompt injection, data poisoning, model manipulation, credential theft, supply chain.
9. **API Standards** — OpenAPI-first REST (/api/v1/), response envelopes, health endpoints, X-Request/Correlation/Trace-ID propagation.
10. **Code Generation Framework (CGF-0001)** — spec compiler that generates OpenAPI, schemas, service scaffolds, SDK stubs, Helm/Terraform, and conformance tests (target: 80% auto-generation, 70% dev-time reduction).

Gate B (Business Case + Architecture Approval): RA-0001...RA-0008 equivalents baselined; canonical data model frozen; ADR repository live.

5. Phase C — Standards & Governance Foundation (Weeks 1–12, parallel track)

Objective: Establish the authority everything else depends on. Without RFC-0001, developers cannot build.

Week	Deliverable
1–2	AEP-X Constitution v1.0 + Book of Laws (10 laws); Foundation Charter; governance bodies defined
3–7	RFC-0001 Core Protocol, RFC-0002 Identity & Trust, RFC-0003 Messaging/Events/Workflow, RFC-0004 Discovery & Federation, RFC-0005 Memory/Knowledge/Cache — frozen as v1.0 Draft
6	GitHub organisation + monorepo (<code>aepx-core</code> : RFCs, schemas, SDKs, services, tests, docs); CI/CD; contribution guidelines
8–12	RFC-0006 Governance/Compliance/Audit, RFC-0007 Safety/Anti-Hallucination, RFC-0008 Economic Optimisation, RFC-0009 Marketplace, RFC-0010 Certification

Standards lifecycle: Draft → Review → Public Consultation → TSC Approval → Publication → Deprecation. No breaking changes within v1.x (SEMVER discipline).

Governance bodies (minimum viable): Technical Steering Committee, Security Council, Certification Council, Community Council; Board-level Audit & Risk Committee. Full Foundation registration targeted Month 18.

Gate C (Standards Freeze): RFC-0001–0005 published and frozen; monorepo and CI/CD operational.

6. Phase D — Implementation (Months 1–12)

Objective: Build the 13-service core runtime plus SDKs, cache-first, spec-driven. This is the manual's 180-day Execution Lock scope followed by ecosystem waves.

6.1 Build order (critical path)

The manual's critical path is strictly sequential because of the constitutional laws (identity before trust, trust before execution):

Wave 1 – Access Layer: Registry → Discovery → Gateway
Wave 2 – Intelligence Layer: Memory → Cache → Knowledge
Wave 3 – Trust Layer: Identity → Trust → Policy → Audit
Wave 4 – Execution & Safety: Workflow → Safety → Verification
Wave 5 – Ecosystem: Marketplace → Certification → Developer Portal → Billing → Licensing
Wave 6 – Platforms & Pilots: Cloud MVP → Education → Hospitality → Enterprise

Note: identity/trust plumbing (JWT, OIDC) is stubbed in Wave 1 and hardened in Wave 3 — the manual's week-by-week plan builds Identity and Trust first in minimal form (Week 1) and productionises them in Wave 3.

Detailed build reference: the bounded-context service boundaries, communication patterns (sync Tier 1 vs. event-driven Tier 2), database-per-service enforcement, and step-by-step code for Wave 1–4 are specified in the companion **Microservices Implementation Guide**, which also reconciles this Wave numbering against the AEP-X Ultra book outline's Volume 9 service catalogue (Agent Registry, Trust Authority, Discovery, Memory, Workflow Engine, Safety Engine, Governance Engine, Marketplace Engine).

6.2 Month-by-month schedule (Year 1 Operating Plan)

Month	Deliverables	Milestone
1	Identity + Trust (minimal), Registry, Memory, Cache, Gateway MVPs; Python SDK; Tutor Agent demo	v0.0.1 — first installable version (docker compose up)
2	Discovery Service, Workflow Engine, Vector Memory (pgvector), MCP compatibility adapter	Runtime Kernel (execution, scheduling, routing)
3	Knowledge, Evidence, Anti-Hallucination, Policy services; TypeScript SDK; CLI (<code>aepx init/create/run/deploy</code>)	v0.1 Alpha
4	Universal Connectivity Layer: protocol adapters (MCP/REST/GraphQL/gRPC/Kafka/MQTT), AI connectors (OpenAI, Anthropic, Gemini, Ollama, vLLM...), first tool/data connectors	Connect-once-use-everywhere proven
5	Hardened Identity/Trust/Policy/Audit (Wave 3 complete); observability stack (Prometheus, Grafana, OpenTelemetry, Jaeger, OpenSearch)	Trusted execution, 100% audit coverage
6	Marketplace MVP, Trust Authority, Cloud MVP; load/security/compliance testing	v0.5 Beta → v1.0 RC
7–9	Education pilot (Tutor/Curriculum/Assessment agents), Hospitality pilot (Search/Booking/Pricing), first enterprise pilot; case studies; beta programme	3 pilots running
10–12	Certification platform, Developer Portal, Enterprise Edition packaging; GA hardening; documentation portal	v1.0 GA (Month 12)

6.3 Engineering practices

- **Spec-driven development:** services generated/validated against RFC schemas via CGF-0001; 100% standards compliance and API consistency are hard requirements.
- **Sprint cadence:** 2-week releases (Engineering Board KPI); every sprint follows the manual's 20-step structure ending in explicit success criteria.

- **Team structure (Year 1, ~7 people):** Founder/Chief Architect; 2× Backend Engineers; AI Engineer; DevOps Engineer; QA Engineer; Product Owner — organised as Team A (Core Platform), Team B (Developer Experience), Team C (Trust & Security), Team D (Pilots). Growth: 12–15 (Year 2), 20–25 (Year 3).
- **Definition of Done (every service):** OpenAPI contract + schema published; unit tests >85–90% coverage; integration tests for critical flows; Dockerfile + Helm chart; observability instrumented; audit events emitted; conformance tests passing; docs published.

Gate D (MVP Approval): All 13 core services operational; the full chain Gateway → Registry → Discovery → Workflow → Safety → Verification executes end-to-end.

7. Phase E — Validation (continuous; formal gates at Months 3, 6, 12)

7.1 Test categories (RFC-0010 conformance suite)

Category	Scope	Key thresholds
Functional	Protocol, APIs, workflows, memory, trust, safety	100% critical flows
Performance	Latency, throughput, scaling	See §9 targets
Security	AuthN/AuthZ, encryption, audit, dependencies, secrets	0 critical vulnerabilities
Safety	Evidence coverage, verification, risk classification, hallucination testing	≥95% evidence coverage, <1% critical hallucination on 1,000-question benchmark
Compliance	GDPR, UK GDPR, EU AI Act, ISO 27001/42001, SOC 2	100% of controls evidenced
Interoperability	MCP compat, federation, cross-org workflows, multi-region	3 independent implementations (GA+)

7.2 Alpha Gate Review (Month 3) — five gates

1. **Technical:** Discovery <50 ms, Memory <20 ms, Cache >90% hit, Gateway <100 ms.
2. **Developer Experience:** install <5 min, first agent <10 min, first workflow <15 min, deployment <30 min.
3. **Cost:** measured cost per execution vs plain-LLM baseline (target trajectory to 80% LLM-call reduction).
4. **Anti-Hallucination:** 100-question known-answer benchmark; evidence and confidence scoring validated.
5. **Competitive Benchmark:** vs MCP, LangGraph, CrewAI, AutoGen, OpenAI Agents SDK on latency, cost, complexity, extensibility, DX.

Decision: refactor or proceed to Beta. The parallel **Reality Validation Programme** publishes the benchmark report (pass/fail per area) — this is the honesty mechanism against over-engineering.

7.3 Beta exit (→ RC) and RC exit (→ GA)

- **Beta exit:** 99% reliability; <100 ms gateway; <20 ms memory; 3 pilots; 100 developers; 10 organisations; measured 50% cost reduction.
- **RC exit:** 99.9% reliability; 90% cache hit; 80% LLM reduction; conformance suite published; independent security audit passed; 3 production pilots; 1,000 developers; positive unit economics; funding secured.

8. Phase F — Release Management

Versioning: SEMVER (PATCH = fix, MINOR = feature, MAJOR = breaking). v1.x backward compatible. Python & TypeScript SDKs supported ≥3 years.

Release	Target	Content	Exit criteria owner
v0.0.1	Month 1	6 minimal services + Python SDK + demo agent	Engineering Board
v0.1 Alpha	Month 3	+ Discovery, Workflow, Vector Memory, MCP compat, Anti-Hallucination	Alpha Gate Review
v0.5 Beta	Month 6	+ Security hardening, UCL connectors, Marketplace MVP, cost optimiser, pilots	Beta exit gates
v1.0 RC	Months 6–9	Core frozen; specs published; conformance suite; security audit	RC exit gates
v1.0 GA	Month 12	Full platform + 3 verticals + Certification + Foundation launch	GA launch programme (10 deliverables)
Post-GA	Months 13–24	Enterprise/Government editions; LTS programme; ecosystem scale	PEB

GA Launch Programme (10 deliverables): public launch (site/portal/forum), Foundation formalised, certification programme live, marketplace launch (targets: 100 agents, 50 workflows, 20 plugins at launch; 1,000/500/100 within first release cycle), 10 university partnerships, industry accelerators, published benchmarks, commercial model (Open Source / Professional £99–499/mo / Enterprise £25k–250k/yr), standards expansion working groups, AEP-X 2.0 research kickoff.

9. Non-Functional Targets (binding through all phases)

Performance & reliability

Metric	MVP (Mo. 6)	GA (Mo. 12)	Year 3
Availability	99.9%	99.99%	99.999%
Gateway latency	<100 ms	<100 ms	<100 ms
Discovery	<50 ms (<10 ms cached)	<50 ms	<50 ms
Memory retrieval	<20 ms (<5 ms cached)	<20 ms	<20 ms
Cache hit lookup	<5 ms	<5 ms	<5 ms
Policy / trust evaluation	<50 ms	<50 ms	<50 ms
Audit write / search	<10 ms / <100 ms	same	same
Workflow start / planning / recovery	<100 / <200 / <500 ms	same	same
Verification per claim	<200 ms	<200 ms	<200 ms

Economics (the platform's core differentiator)

Metric	Target
Cache hit rate	>90% (→95%)
LLM call reduction	80% (→90%)
Memory reuse	90%
Workflow reuse	70–80%
Infrastructure cost reduction	50%
Energy reduction	30% (→50% long-term)

Safety & quality

Metric	Target
Evidence coverage	≥95%
Verified responses	≥90%
Critical hallucination rate	<1%
Human oversight, S3/S4 & regulated domains	100%
Auditability / traceability / explainability	100%
Unit test coverage	>85–90%
Defect escape rate	<2%
Release success rate	>95%

10. Compliance & Certification Roadmap (parallel track, Months 1–30)

Phase	Months	Outcome
1	1–6	Governance foundation: POLICY-001, RISK-001, QUALITY-001, DATA-001, AUDIT-001 live
2	6–12	ISO 9001 (QMS)
3	9–15	Cyber Essentials Plus
4	12–18	ISO 27001 (ISMS)
5	15–21	ISO 22301 (BC/DR: RTO <4 h, RPO <1 h for Tier-1 services)
6	18–24	ISO 42001 (AI management) — supported by AIA-001 AI Assurance Framework
7	24–30	SOC 2 readiness

AI-specific controls (AIA-001, EU AI Act): risk classification R0–R4 for every AI system; DPIAs for all high-risk features; AI system inventory; transparency metadata (purpose, owner, version, risk level, limitations); explainability contract on every response ({answer, confidence, evidence[], verification_status}); bias & fairness monitoring; mandatory human oversight for high/critical risk. Data retention: audit logs 7 years, security logs 2 years, application logs 90 days.

Incident response: P1 15-min / P2 1-h / P3 4-h / P4 next release. Vulnerability SLAs: critical 24 h, high 7 d, medium 30 d.

11. Governance, Risk & Decision Gates

Stage-gate framework (PMO-001)

Gate	Name	When
1	Concept Approval	End of Phase A
2	Business Case Approval	End of Phase B
3	MVP Approval	Month 6
4	Pilot Approval	Months 7–9
5	Commercial Launch Approval	Month 12 (GA)
6	Expansion Approval	Month 18–24

Programme Execution Board (weekly command centre)

Six boards with single KPIs: Standards (RFCs frozen/traceable), Engineering (release every 2 weeks), Security (zero critical vulnerabilities), Product (one new pilot per quarter), Commercial (first paying customer, then ARR), Foundation (100 active contributors). Five dashboards: Product (WAU — *the single most important KPI*), Technology, Economics, Trust & Safety, Growth.

Top risks & mitigations

Risk	Score	Mitigation
Data breach	Critical (20)	Encryption everywhere, MFA 100%, monitoring, security reviews
GDPR non-compliance	Critical (16)	Privacy by design, DPIAs, training, DATA-001
Funding delays	High (15)	Staged plan: bootstrap/grants (£250k, Mo. 6) → seed £1M+ (Mo. 24) → Series A (Yr 3–4); Innovate UK/UKRI applications in parallel from Month 1
Supply chain attack	High (15)	SBOM, dependency scanning, code signing (Trivy/Syft/Cosign)
Slow adoption	High (12)	Pilot early, university partnerships, free open-source core, <10-min first agent
EU AI Act non-conformity	High (12)	AIA-001 framework, independent reviews
Over-engineering	High	180-day Execution Lock ; Reality Validation Programme; WAU as north-star metric
Vendor capture / fragmentation	Medium	Non-profit foundation, open RFCs, conformance testing, 3 independent implementations

12. Phase G/H — Operations, Learning & Evolution (Month 6 onward, permanent)

- **Operations (OPS-001):** GitOps deployment (ArgoCD, Helm, Terraform); multi-AZ; automated backups (hourly incremental, daily full, 35-day retention); quarterly tabletop exercises, twice-yearly recovery tests, annual full DR exercise; SLO-based alerting.
- **Learning loop (Law 9):** every interaction feeds Experience → Knowledge → Learning → Improvement; the ML fabric (classification, prediction, recommendation) retrains from outcomes; cost optimiser continuously re-tunes model routing.

- **Evolution loop (Law 10):** telemetry and pilot feedback drive RFC updates through the standards lifecycle; AEP-X 2.0 research stream starts at GA; deprecation policy protects backward compatibility.
- **Post-GA programme:** independent implementations, interop lab, conformance certification, connector expansion, global education programme, industry accelerators (Education, Hospitality, Government, Healthcare, Finance), AEP-X Cloud.

13. Adoption & Commercial Milestones (success criteria per period)

Milestone	Developers	Organisations	Universities	Revenue	Release
Month 3	100	1–3 pilots	1 contact	—	v0.1 Alpha
Month 6	500	10	1–2	first paying customer	v0.5 Beta / RC
Month 12	1,000	100	10	£100k ARR	v1.0 GA
Month 24	5–10k	500–1,000	25–30	£500k–£1M ARR	Ecosystem
Year 3	5,000–10,000	200–1,000	25–100	£2M ARR	Enterprise scale
Year 5	100,000	1,000	100	£10M+ ARR	Industry leadership

Marketplace at first release cycle: 1,000 agents, 500 workflows, 100 plugins, 100 knowledge packs. Certification Year 1: 100 certified developers (5-level track: Developer → Engineer → Architect → Enterprise Architect → Fellow; exams £150–£1,500).

14. Immediate Next Actions (Days 1–30)

1. **Week 1:** Register GitHub org + domains; publish Constitution v1.0 draft and Book of Laws; stand up docs portal skeleton.

2. **Week 2:** Freeze RFC-0001–0005 as v1.0 Draft; publish OpenAPI schemas and canonical data model; monorepo + CI/CD live.
3. **Week 3:** First running services — Identity (minimal), Trust (minimal), Registry, Memory, Cache, Gateway via docker compose.
4. **Week 4:** Python SDK v0.0.1 (`pip install aepx`); Tutor Agent demo end-to-end; measure: time-to-first-agent <10 min.
5. **In parallel:** Innovate UK application drafted; 2 university conversations opened; risk register and policy library (POLICY-001) initialised.

Definition of success for Month 1: a new developer can go from `git clone` to a working evidence-validated agent in under 10 minutes — everything else in this plan exists to keep that loop true at scale.

15. Critical Evaluation

This section assesses the manual — and therefore this plan — honestly. It exists because the manual itself demands it: the Reality Validation Programme, the 180-day Execution Lock, and the warning that "*if nobody uses AEP-X, architecture = 0 value*" are all the manual auditing its own ambition. The evaluation below extends that audit.

Overall verdict: The manual is a visionary and unusually complete *portfolio of ambitions*, but it is not yet an executable plan. Its genuine differentiators (cache-first economics, evidence-first anti-hallucination, compliance-by-design) are buried under a scope that no Year-1 team of 7 can deliver, and the document carries material internal contradictions from iterative drafting that must be reconciled before build. This ADLC plan is executable **only under the manual's own Execution-Lock reading**: 6 services, one SDK, one vertical, one paying customer — with everything else held as conditional backlog.

15.1 What is genuinely strong

1. **Cache-first economics is a real, measurable differentiator.** The execution hierarchy (cache → memory → knowledge → ... → LLM last) attacks the dominant cost line of agentic systems, and no mainstream framework (MCP, LangGraph, CrewAI, AutoGen) makes it a protocol invariant. It is also falsifiable — the cost dashboard either shows the savings or it doesn't.
2. **Evidence-first anti-hallucination fills a genuine gap.** A response contract of `{answer, confidence, evidence[], verification_status}` with risk-tiered human oversight is exactly what regulated buyers (education, healthcare, government) ask for and don't get from current stacks.
3. **Compliance-by-design is well timed.** Building EU AI Act / ISO 42001 conformity into the runtime, rather than bolting it on, matches where enterprise and public-sector procurement is heading in 2026–2028.
4. **The manual is self-aware.** The Execution Lock, the Reality Validation Programme, the "WAU is the only KPI that matters" directive, and the benchmark-vs-competitors gates are the document correcting its own tendency to sprawl. The best strategic thinking in the manual is its scepticism about itself.
5. **The critical path is architecturally sound.** Identity → trust → registry → memory → execution is the right dependency order, and the 20-step sprint template with explicit success criteria is a solid delivery discipline.

15.2 Critical weaknesses

1. **Scope–capacity mismatch is the central defect.** Year 1 asks a team of 7 to deliver 13 production services (>85% test coverage, Helm, observability, audit), 2 SDKs, a CLI, a marketplace, a certification platform, 3 vertical pilots, a standards body, 10 RFCs, and 2 ISO certifications. That is roughly 2 production services per engineer *plus* everything else. Kubernetes had Google behind it; MCP has Anthropic. At production quality this is a 25–40 person scope. The month-by-month schedule in §6.2 is achievable only at MVP/scaffold quality.

2. **Head-on competition with MCP is the largest market risk.** The manual positions AEP-X as an MCP superset while MCP has first-mover network effects and a major backer. A protocol war fought by a bootstrapped startup is unwinnable on those terms; an MCP-*complementary* position (the memory/cache/trust/evidence layer that makes MCP deployments cheaper and safer) is winnable.
3. **The standards-body strategy is inverted.** Constitution, Book of Laws, Foundation, and 10 frozen RFCs are scheduled *before* a single external user exists. Every successful precedent (TCP/IP, OAuth, Kubernetes/CNCF) ran the other way: working implementation → adoption → formal standardisation. Freezing standards with zero implementers creates prestige overhead, not authority.
4. **Key performance targets are asserted, not evidenced.** See the reality-check table (§15.4). The 90–95% cache-hit target — from which the 80% LLM-cost-reduction claim mechanically follows — assumes highly repetitive workloads; real agentic traffic is long-tailed. Publishing these as universal platform guarantees before pilot data exists risks a credibility failure at exactly the moment the benchmark reports (Alpha Gate 5) go public.
5. **Financial and adoption figures contradict each other** (§15.3) and the funding maths does not close: £250k of grant/angel funding cannot carry 7 UK FTEs for 12 months (~£600–700k loaded cost). Either the team is 3–4 people, or more capital is needed earlier — the plan currently pretends neither.
6. **No adoption mechanism behind the adoption targets.** 1,000 developers by Month 12 is asserted with no DevRel headcount, no marketing budget line beyond "community", and no distribution wedge other than open-source virality — against an incumbent with default distribution.
7. **Compliance overhead front-loaded onto scarce capacity.** Six ISO certifications in 30 months while building the platform diverts the same 7 people. Only GDPR fundamentals and Cyber Essentials are needed before the first enterprise deal; ISO 42001 and SOC 2 matter post-revenue.
8. **The manual's provenance shows.** The 3,188-page consolidated edition stacks ~20 successive re-plannings of the same programme (Stages, Phases, G-phases, waves, sprints, 100-day/90-day/30-day plans) without reconciliation. Terminology drifts between passes (§15.3). Anyone implementing directly from the manual, rather than from a reconciled plan like this one, will build inconsistent systems.

15.3 Contradiction register (must be resolved before build)

Topic	Conflicting statements in the manual	Resolution adopted in this plan
Memory layers	M0–M5 (6 layers, p.11) vs M0–M6 (7 layers, RFC-0005)	M0–M6 (RFC-0005 is the normative source)
Cache layers	L1–L4 vs L0–L5 vs L0–L6 in different chapters	L0–L5 (RFC-0005)
Availability target	99.9% vs 99.99% vs 99.999% quoted for overlapping periods	Staged: 99.9% MVP → 99.99% GA → 99.999% aspirational Year 3
Trust score formula	5 equal components × 20% (RFC-0002) vs 40% reputation / 30% compliance / 20% reliability / 10% security (build chapters)	RFC-0002 formula; revisit with pilot data
Safety classes	S0–S3 (RFC-0007) vs S0–S4 (multiple chapters)	S0–S4, human oversight mandatory at S3+
RFC numbering	RFC-0008 = "Safety Framework" (Stage 2) vs RFC-0008 = "Economic Optimisation" (RFC suite); "8 core standards" vs "10 RFCs"	10-RFC suite as listed in §5
Year-1 adoption	100 devs / 10 orgs (EXEC-001) vs 1,000 devs / 100 orgs (Y1OP GA criteria) vs 10,000+ devs / 1,000+ orgs (ROADMAP-001 "Year 1")	Plan uses the Y1OP ladder (§13); treat EXEC-001 numbers as the commit case, Y1OP as stretch
Year-1 revenue	£50k–£100k (FIN-001) vs £100k ARR (Y1OP) vs ">£400k" (investor KPIs)	£100k ARR stretch, £50k commit
MVP duration	90 days vs 6 months vs 9 months vs 12 months across programmes	v0.0.1 at 30 days, Alpha at 90 days, GA at 12 months

15.4 Reality check on headline targets

Claim	Assessment
>90–95% cache hit rate	Conditional. Plausible for narrow, repetitive domains (FAQ-style education pilots); typical agentic long-tail workloads land far lower. Should be an SLO per workload class, never a platform-wide guarantee.
80% LLM call/cost reduction	Derivative of the above. Credible in the education pilot; unproven generally. Publish measured pilot numbers, not the target.
<20 ms memory retrieval	Optimistic for semantic search. pgvector ANN at scale typically runs 20–100 ms+; achievable only for cache-fronted reads. Keep <5 ms cached / restate uncached honestly.
99.999% availability (Year 3)	Conflicts with the low-cost positioning. Five nines means multi-region active-active and a serious SRE bench — a >£1M/yr operations posture. 99.9→99.95% is the honest Year 1–3 band.
13 services in 6 months, team of 7	Not credible at production grade. Feasible only as code-generated MVP scaffolds — and the code generator (CGF-0001) is itself an unbuilt dependency on the critical path.
<1% critical hallucination, ≥95% evidence coverage	Unmeasurable as stated. Needs a defined benchmark methodology (dataset, judge protocol, domain spread) before the number means anything. The 1,000-question benchmark is a start; pre-register it.
1,000 developers by Month 12	No mechanism. Requires funded DevRel and an MCP-compatible on-ramp; treat 100–300 as the commit case.
Standards-body recognition (IETF/W3C/CNCF peer)	Years premature. Recognition follows adoption; schedule after 3 independent implementations exist, not before.

15.5 Recommendations

1. **Make the Execution Lock the actual plan.** Year-1 committed scope: 6 services (Identity, Trust, Registry, Memory, Cache, Gateway), Python SDK, education vertical, 1 paying customer. Waves 5–6, the Foundation, and 5 of the 10 RFCs move to a conditional backlog gated on pilot evidence.

2. **Reposition relative to MCP: complement, not competitor.** Ship the MCP adapter in Month 1, not Month 4, and market AEP-X as "the memory, cost and evidence layer for MCP agents". Inherit the ecosystem instead of fighting it.
 3. **Demote RFCs to living design docs** until there are third-party implementers; freeze only the wire format and response contract. Register the Foundation at Month 18 as planned, not earlier.
 4. **Fix the funding maths before hiring.** Either raise ~£600–750k pre-seed for the 7-FTE plan, or run Months 1–6 with 3–4 people and cut the Month-6 milestone list accordingly. The current £250k/7-FTE combination fails in Month 5.
 5. **Re-baseline public targets to defensible numbers** (99.9% availability; cache-hit by workload class; hallucination metrics only after the benchmark methodology is pre-registered). Keep aspirational numbers internal until pilots evidence them.
 6. **Build the public benchmark suite (RVP) in Month 2, not after Alpha.** It is the credibility engine for every economic claim the project makes — and the cheapest marketing available.
 7. **Adopt the contradiction register (§15.3) as a living document** with a single normative source per topic; every spec/service PR links to it.
 8. **Defer ISO 42001 and SOC 2 to post-revenue;** keep GDPR fundamentals, Cyber Essentials, and the AI risk-classification practice from day one because the anti-hallucination product story depends on them.
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Appendix A — Reconciliation Map (manual roadmaps → this plan)

Manual construct	Where it lands in this ADLC
SPARC / SPARC-X lifecycle	§2 methodology; phases A–H
Development Guide Phases 1–20	Phases A–E deliverables
Genesis Implementation Programme G1–G10	G1–G2 → Phase C; G3–G4 → Phase D; G5–G7 → Waves 5–6 + §13; G8 → §11 funding; G9–G10 → §12 post-GA
Stage 5 (12 workstreams, 9 months) & Stage 14-bis (12-month build)	§6.2 month-by-month schedule
100-Day Plan / 90-Day FES / Day 1–30 checklist	§14 immediate actions + §6.2 Months 1–3
24-Month Execution Plan (EXEC-001)	§6.2 + §13 milestones
MVP v1.0 6-phase / 6-sprint plans	§6.1 build waves
Alpha/Beta/RC/GA programmes	§7 gates + §8 releases
ISO-001 certification roadmap	§10
TRANSFORM-001 5-year / SEP 10-year	§13 (Years 3–5 summarised; out of ADLC execution scope)
RVP / PoV / FCP / SRP / SGSG	§7.2 validation, §12 evolution, §13 adoption

Appendix B — Technology Stack (reference implementation)

FastAPI (services) · PostgreSQL + pgvector · Neo4j · Redis/Redis Cluster · Kafka · Temporal (workflow) · Keycloak (identity) · Open Policy Agent · HashiCorp Vault · Docker/Kubernetes/Helm/Terraform/ArgoCD · Prometheus/Grafana/OpenTelemetry/Jaeger/OpenSearch/AlertManager · MinIO · Next.js (portals) · Python & TypeScript SDKs (then Java, Go, Rust).